



GREEN TECHNOLOGY

INDIA'S DEFENCE SECTOR

FOCUSING ON INDIGENOUS AND GREEN TECHNOLOGIES



"I fear the day that technology will surpass our human interactions. The world will have a generation of idiots." - Albert Einstein

The Paris Agreement on climate change is now the basis for many decisions on the manufacturing processes and systems that should be followed all over the world. While industry at large slowly moves toward the implementation of green technology, international militaries cannot be left behind in the race to adopt this technology. Certain ambitious targets have been set by the Paris Agreement, in terms of lowering energy consumption, reducing our dependence on fossil fuels and slashing carbon emissions. Across all verticals, the adoption of more environment-aware manufacturing processes is referred to as green manufacturing.

Today, India's armed forces are the third largest in the world and their combined expenditure is the eighth largest defence budget, globally. In 2016, on World Environment Day, the broad

outline of how to reduce the carbon footprint of the Indian armed forces was chalked out. The Indian Army recently created an energy and environment cell, which is tasked with making it an environmentally responsible force – not only energy and resource efficient, but resilient in terms of energy costs and disruptions. The main aims of this mission are energy conservation, diversification of energy supplies and minimising the impact on the environment.

These aims are being achieved by using cutting-edge technology and state-of-art equipment, which are energy efficient and have a low carbon footprint. Mass energy conservation drives have become the norm, driven by this cell. Regular mentoring sessions are held across the commands and repair stations, to continuously educate and sensitise personnel on the importance of energy

and environmental conservation, and to make environment consciousness a way of life. These initiatives have borne fruit, with significant savings to the tune of Rs 15 million to Rs 200 million annually, per naval base station.

The Defence Institute of Bio-Energy Research (DIBER), a DRDO lab based at Haldwani in Uttarakhand, has developed a bio-diesel from a plant called *Jatropha Curcas*, which is a blend that includes 20 per cent normal diesel. Trials for this have been successful on ships like the *INS Shivaji*. And now, with the inclusion of the bio-diesel powered fast interceptor combat vehicles, the progress in this new direction is encouraging.

An overview of the changes

India today accounts for 14 per cent of global arms imports (see Figure 1 for global details), and 2.4 per cent of the

